The three-dimensional core–redundant case

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Consider

$$X \sim N_3(\mu, I_3) \ ,$$

where coordinates 1 and 2 are core moments and coordinate 3 is redundant. The maintained mean space is

$$\Lambda := \left\{\mu \in \mathbf{R}^3 : \mu_3 \leq \max\{\mu_1, \mu_2\}\right\} \ .$$

The null and alternative are

$$\Lambda_0 := \Lambda \cap (-\infty, 0]^3, \qquad \Lambda_1 := \{\mu \in \Lambda : \max\{\mu_1, \mu_2\} > 0\} \ .$$

Fix $0 < \alpha < 1$. A level-α test is called α-admissible if there is no other level-α test whose power is weakly larger at every $\mu \in \Lambda_1$ and strictly larger at some $\mu \in \Lambda_1$.

The test of primary interest is the equal-cutoff rectangle, or max test. Its acceptance region is

$$A_\kappa := \bigcap_{j=1}^{3} \{x_j \leq \kappa\} \ , \tag{1}$$

and its rejection rule is

$$\phi_{\max}(x) = I\left\{\max_{1\leq j\leq 3} x_j > \kappa\right\} \ .$$

Thus the statistic is the maximum of all three coordinates and a single critical value is applied to that maximum. Under independence, exact size at the origin requires

$$\Phi(\kappa)^3 = 1 - \alpha, \qquad \kappa = \Phi^{-1}\{(1-\alpha)^{1/3}\} \ .$$

Coordinatewise lowerness then makes the test level α over Λ_0, with size attained at $\mu = 0$.

1 Failure of the feasible-normal sufficient condition

For the present identity-covariance experiment, the feasible-normal set in Proposition 1 of the extension plan becomes

$$\mathcal{F}(I_3, \Lambda) = \left\{a \in \mathbf{R}^3 : a \geq 0, \quad a_3 \leq \max\{a_1, a_2\}\right\} \ . \tag{2}$$

Indeed, the general condition is $a \geq 0$ and $I_3 a = a \in \Lambda$.

The support function of the rectangle in (1) is

$$h_{A_\kappa}(a) = \begin{cases} \kappa(a_1 + a_2 + a_3), & a \geq 0, \\ +\infty, & \text{otherwise.} \end{cases}$$

Consequently, every nonzero direction $a \geq 0$ is a finite-support direction of A_κ. In particular, the normal to the redundant-coordinate face is

$$e_3 = (0, 0, 1), \qquad h_{A_\kappa}(e_3) = \kappa < \infty \ .$$

But

$$(e_3)_3 = 1 > \max\{(e_3)_1, (e_3)_2\} = 0 \ ,$$

so $e_3 \notin \mathcal{F}(I_3, \Lambda)$. The equal-cutoff rectangle therefore fails Proposition 1 of the extension plan.

The same conclusion holds for every finite three-dimensional rectangle, not only for the equal-cutoff rectangle, because each such rectangle has the finite normal e_3. This calculation does not establish that the max test is inadmissible. The extension-plan result is sufficient, not necessary, and its failure leaves the admissibility of the rectangle unresolved.

2 Limits of the two-dimensional gated repair

In the two-dimensional problem with one fixed core moment, a rectangular face for the redundant moment can be replaced by a calibrated gate involving the sum of the core and redundant statistics. The power comparison works because every alternative satisfies both

$$\mu_c > 0 \qquad \text{and} \qquad \mu_3 \le \mu_c$$

for the same fixed core coordinate c.

There is no such fixed coordinate in the present model. The alternative is the union of the two branches

$$\Lambda_1^{(1)} := \{\mu : \mu_1 > 0,\ \mu_3 \le \mu_1\}, \qquad \Lambda_1^{(2)} := \{\mu : \mu_2 > 0,\ \mu_3 \le \mu_2\} \ ,$$

where the unpaired core mean is unrestricted on each branch. On the first branch, the likelihood-ratio exponent can be written as

$$\mu' x = \mu_1(x_1 + x_3) + (\mu_1 - \mu_3)(-x_3) + \mu_2 x_2 \ . \tag{3}$$

The first two coefficients in (3) are nonnegative in the directions required by the bivariate transfer argument. This supports a gate pairing coordinates 1 and 3, while coordinate 2 is left unchanged. The analogous decomposition on the second branch supports a gate pairing coordinates 2 and 3. A gate based on one fixed core coordinate need not improve power on the other branch: that selected core mean may be negative and may lie below μ_3.

A natural attempt to avoid selecting a core coordinate in advance is the symmetric gated acceptance region

$$A_b^{\mathrm{g}} := \{x_1 \le \kappa,\ x_2 \le \kappa,\ x_1 + x_3 \le b,\ x_2 + x_3 \le b\} \ . \tag{4}$$

Equivalently, the redundant face is replaced by

$$x_3 + \max\{x_1, x_2\} \le b \ .$$

Choose b so that

$$P_0(A_b^{\mathrm{g}}) = P_0(A_\kappa) = 1 - \alpha \ .$$

The finite facet normals of A_b^{g} are

$$e_1, \quad e_2, \quad (1,0,1), \quad (0,1,1) \ ,$$

and each belongs to (2). This alone is not enough for the extension-plan condition because the feasible-normal set is not convex and that condition concerns every finite-support direction. In particular, the sum of the two gated-face normals is $(1,1,2)$. It has $h_{A_b^{\mathrm{g}}}(1,1,2) = 2b < \infty$, but is not feasible because $2 > \max\{1,1\}$. Thus the symmetric gated test, like the original rectangle, falls outside the sufficient condition. More importantly here, it does not dominate the original rectangle.

In fact, the symmetric gate does not dominate. Its acceptance probability at a general mean is

$$P_\mu(A_b^{\mathrm{g}}) = \int_{-\infty}^{\infty} \varphi(z-\mu_3)\Phi(\min\{\kappa, b-z\}-\mu_1)\,\Phi(\min\{\kappa, b-z\}-\mu_2)\ dz \ , \tag{5}$$

whereas

$$P_\mu(A_\kappa) = \prod_{j=1}^{3} \Phi(\kappa - \mu_j) \ . \tag{6}$$

For example, take $\alpha = 0.05$. Null calibration gives

$$\kappa \simeq 2.121201, \qquad b \simeq 3.117267 \ .$$

At

$$\mu = (0.1, -5, 0.1) \in \Lambda_1 \ ,$$

equations (5) and (6) give

$$P_\mu(A_\kappa) \simeq 0.957209, \qquad P_\mu(A_b^{\mathrm{g}}) \simeq 0.965438 \ .$$

The corresponding rejection probabilities are approximately 0.042791 and 0.034562. The symmetric gated test therefore has lower power at this alternative.

The example exposes the adaptation cost created by the maximum restriction. When μ_2 is very negative, coordinate 1 is effectively the only relevant core coordinate. Nevertheless, the symmetric gate is calibrated under the null to allow either core coordinate to activate the gate. The resulting critical value is too large in this nearly bivariate subproblem.

There is also a geometric obstruction to a simple symmetrization. A sample-space move d that increases every alternative likelihood ratio would have to satisfy

$$\mu' d \geq 0 \qquad \text{for every } \mu \in \Lambda_1 \ .$$

The unrestricted value of μ_2 on $\Lambda_1^{(1)}$ forces $d_2 = 0$, and the unrestricted value of μ_1 on $\Lambda_1^{(2)}$ forces $d_1 = 0$. With these equalities, taking $\mu_3 = \mu_1 > 0$ on the first branch gives $d_3 \geq 0$, while allowing $\mu_3 \to -\infty$ gives $d_3 \leq 0$. Hence the common likelihood-ratio transfer cone is

$$\{0\} \ .$$

There is no nonzero pointwise move that improves every alternative in both branches. Any dominating test, if one exists, must therefore use a genuinely distributional argument that adapts to the information in the data about which core coordinate is active. This observation rules out the direct transfer argument, but it does not determine admissibility. The next two sections give sharper sufficient conditions that establish admissibility of both the symmetric gate and the rectangle.

3 A sharper sufficient condition based on defining supports

The sufficient condition in Section 1 requires every direction in which the acceptance region has finite support to be feasible. Fabel's observation is that this is stronger than needed in the present two-core model. It is enough that the feasible supporting halfspaces already determine the acceptance region.

Call an acceptance region A *nice* if it is closed, convex, coordinatewise lower, and satisfies $P_0(A) = 1 - \alpha$. Let

$$\mathcal{F} := \{a \in \mathbf{R}^3 : a \geq 0,\ a_3 \leq \max\{a_1, a_2\}\}$$

and define

$$F_A := \{a \in \mathcal{F} \setminus \{0\} : h_A(a) < \infty\} \ .$$

We say that A is *feasibly supported* if

$$A = \bigcap_{a \in F_A} \{x : a'x \leq h_A(a)\} \ . \tag{7}$$

For a polyhedral region, this means that its essential facet normals can all be chosen feasible.

Proposition 1 (Feasible support suffices)**.** *Every feasibly supported nice test is α-admissible against all level-α tests, including randomized competitors.*

Proof. Let $\phi_A := I\{X \notin A\}$, suppose that a randomized level-α test ψ weakly dominates ϕ_A, and put $h := \psi - \phi_A$. Since $0 \in \Lambda_0$, level control gives $E_0\psi \leq \alpha = E_0\phi_A$. Dominance along $te_1 \in \Lambda_1$ and continuity as $t \downarrow 0$ give the reverse inequality, so

$$E_0 h = 0 \ . \tag{8}$$

Fix $a \in F_A$. Since a is nonzero and feasible, $a \in \Lambda_1$. Dominance along the ray ta, together with the Gaussian likelihood ratio, gives

$$E_0[e^{ta'X} h(X)] \geq 0 \qquad \text{for every } t > 0 \ . \tag{9}$$

Write $h = h^+ - h^-$. The sign restrictions imposed by ϕ_A give $\{h^+ > 0\} \subseteq A$. If, for some $\varepsilon > 0$,

$$m_{a,\varepsilon} := E_0[h^- 1\{a'X \geq h_A(a) + \varepsilon\}] > 0 \ ,$$

then

$$E_0[e^{ta'X}h(X)] \le e^{th_A(a)}E_0h^+ - e^{t\{h_A(a)+\varepsilon\}}m_{a,\varepsilon} < 0$$

for all sufficiently large t, contradicting (9). Taking $\varepsilon = 1/n$ shows that $h^- = 0$ almost surely on $\{a'X > h_A(a)\}$.

By (7), the open set A^c is the union of these open halfspaces over $a \in F_A$. The Lindelöf property supplies a countable subcover, so the preceding conclusion gives $h^- = 0$ almost surely on A^c. But $\{h^- > 0\} \subseteq A^c$, hence $h^- = 0$ almost surely. Equation (8) then forces $h^+ = 0$ almost surely. Mutual absolute continuity of Gaussian shifts rules out a strict power improvement at any alternative. □

The logical relation with the earlier sufficient condition is

$$\{a \ge 0 : h_A(a) < \infty\} \subseteq \mathcal{F} \quad \Longrightarrow \quad A \text{ is feasibly supported} \quad \Longrightarrow \quad \phi_A \text{ is admissible .}$$

The first implication is not reversible. The old condition controls every finite-support direction, whereas Proposition 1 uses only the supporting halfspaces needed to reconstruct A.

Remark 1 (The symmetric gate is admissible). The four facet normals of A_b^{g} in (4) are e_1, e_2, $(1,0,1)$, and $(0,1,1)$, all feasible. Hence the symmetric gate is admissible by Proposition 1. Its bad finite-support direction $(1,1,2)$ is a sum of facet normals and is not needed to define the region. Thus the symmetric gate and the rectangle are both admissible, while the numerical calculation in Section 2 shows that the symmetric gate does not dominate the rectangle.

4 Admissibility of the rectangular max test

The failure of the feasible-normal condition does not imply inadmissibility in the present model. In fact, the equal-cutoff max test is admissible. The argument does not use equality of the cutoffs and therefore applies to every finite exact-size rectangle.

For finite c_1, c_2, c_3, define

$$A_c := \{x_1 \le c_1,\ x_2 \le c_2,\ x_3 \le c_3\}, \qquad \phi_c := I\{X \notin A_c\} \ . \tag{10}$$

Proposition 2 (Admissibility of finite rectangles)**.** *Suppose that $P_0(A_c) = 1 - \alpha$. Then ϕ_c is α-admissible against all level-α tests, including randomized competitors. In particular, the equal-cutoff max test in* (1) *is α-admissible. More strongly, any level-α test that weakly dominates ϕ_c coincides with it almost surely under every P_μ.*

Proof. Suppose that a randomized level-α test ψ weakly dominates ϕ_c on Λ_1. Define

$$h(x) := \psi(x) - \phi_c(x) \ .$$

Because $\mu = 0$ belongs to the null, level control gives $E_0\psi \le \alpha = E_0\phi_c$. On the other hand, $te_1 \in \Lambda_1$ for every $t > 0$. Dominance along this ray and continuity as $t \downarrow 0$ give $E_0\psi \ge E_0\phi_c$. Consequently,

$$E_0 h = 0 \ . \tag{11}$$

We first show the first-moment identities $E_0[X_j h(X)] = 0$, $j = 1, 2, 3$. Let

$$q := E_0[Xh(X)] \in \mathbf{R}^3 \ .$$

The set Λ_1 is a cone, so $td \in \Lambda_1$ whenever $d \in \Lambda_1$ and $t > 0$. Dominance and the Gaussian likelihood ratio therefore imply

$$0 \le E_{td} h(X) = \exp\{-t^2\|d\|^2/2\} E_0[h(X)\exp\{td'X\}] \ .$$

The expression equals zero at $t = 0$ by (11). Because $|h| \le 1$ and the Gaussian moment-generating function is finite everywhere, differentiation under the expectation is justified in a neighborhood of zero. Differentiating from the right at zero therefore gives

$$d'q \ge 0 \qquad \text{for every } d \in \Lambda_1 \ . \tag{12}$$

For every $s \in \mathbf{R}$, both $(1, s, 0)$ and $(s, 1, 0)$ belong to Λ_1. Hence (12) gives

$$q_1 + sq_2 \ge 0, \qquad sq_1 + q_2 \ge 0 \qquad \text{for every } s \in \mathbf{R} \ ,$$

which forces $q_1 = q_2 = 0$. Moreover, $(1, 0, r) \in \Lambda_1$ for every $r \le 1$. Applying (12) with $r = 1$ and $r = -1$ then gives $q_3 = 0$. We have shown that

$$E_0[X_j h(X)] = 0, \qquad j = 1, 2, 3 \ . \tag{13}$$

It remains to use the sign restrictions imposed by the rectangular test. Write $h = h^+ - h^-$ for the positive and negative parts of h. Since $\phi_c = 0$ on A_c and $\phi_c = 1$ outside A_c,

$$\{h^+ > 0\} \subseteq A_c, \qquad \{h^- > 0\} \subseteq A_c^c \ , \tag{14}$$

pointwise. Along the alternative ray te_j, for $j = 1, 2$, dominance gives

$$E_0\big[e^{tX_j} h(X)\big] \ge 0 \qquad \text{for every } t > 0 \ . \tag{15}$$

We claim that

$$h^- = 0 \text{ a.s. on } \{X_j > c_j\}, \qquad j = 1, 2 \ . \tag{16}$$

To see this, suppose for some $\varepsilon > 0$ that

$$m_{j,\varepsilon} := E_0\big[h^- 1\{X_j \ge c_j + \varepsilon\}\big] > 0 \ .$$

Let $M^+ := E_0 h^+$. The first inclusion in (14) and the preceding display imply

$$E_0\big[e^{tX_j} h(X)\big] \le e^{tc_j} M^+ - e^{t(c_j+\varepsilon)} m_{j,\varepsilon} \ ,$$

which is negative for all sufficiently large t. This contradicts (15). Applying the argument with $\varepsilon = 1/n$, for $n \geq 1$, proves (16).

By (14) and (16), any negative part of h must be supported on

$$\{X_1 \leq c_1,\ X_2 \leq c_2,\ X_3 > c_3\}\ ,$$

whereas the positive part is supported on $\{X_3 \leq c_3\}$. Equation (11) gives $E_0 h^+ = E_0 h^- =: M$. If $M > 0$, these support restrictions imply

$$E_0[X_3 h^+] \leq c_3 M, \qquad E_0[X_3 h^-] > c_3 M\ .$$

The second inequality is strict because $X_3 > c_3$ on $\{h^- > 0\}$ and $E_0 h^- = M > 0$. Consequently, $E_0[X_3 h] < 0$. This contradicts (13). Thus $M = 0$ and $h = 0$ P_0-almost surely. All Gaussian shifts are mutually absolutely continuous, so ψ and ϕ_c coincide almost surely under every P_μ. A strict power improvement is therefore impossible. □

Remark 2 (Why the same proof fails with one core). In the two-dimensional model with one core and one redundant coordinate, the alternative directions can be normalized as $(1, s)$ with $s \leq 1$. If $q = E_0[Xh(X)]$, the local dominance condition becomes

$$q_1 + s q_2 \geq 0 \qquad \text{for every } s \leq 1\ ,$$

which is equivalent to

$$q_2 \leq 0, \qquad q_1 + q_2 \geq 0\ .$$

Thus local dominance does not force $q = 0$. The core ray still rules out negative comparison mass above the core cutoff. Any remaining negative mass lies above the redundant cutoff, which yields $q_2 < 0$. This is consistent with the preceding inequalities and is precisely the direction used by the gated dominator. In the three-dimensional model, the second unrestricted core supplies both $(1, s, 0)$ and $(s, 1, 0)$ for every real s. These directions collapse the permissible first-moment cone to $\{0\}$, converting the same tail comparison into a contradiction.

5 One additional supporting halfspace

The rectangle proof has a broader form. The zero-mass and first-moment arguments in that proof use only the two-core model and exact size, not the shape of the acceptance region. Thus, for any nice A and any weak dominator ψ, with $h := \psi - \phi_A$,

$$E_0 h = 0, \qquad E_0[Xh(X)] = 0\ . \tag{17}$$

The first identity follows by approaching the origin along te_1. For the second, differentiation along $(1, s, 0)$ and $(s, 1, 0)$ for every $s \in \mathbf{R}$, followed by $(1, 0, 1)$ and $(1, 0, -1)$, forces all three components to vanish.

Proposition 3 (Feasible support plus one halfspace)**.** *Let*

$$A = A_g \cap \{x : u'x \le c\}$$

be nice, where $u \neq 0$ is arbitrary and A_g is an intersection of halfspaces with feasible normals. Then ϕ_A is α-admissible.

Proof. Let ψ weakly dominate ϕ_A and define $h := \psi - \phi_A$. For every feasible halfspace defining A_g, the exponential-tilt argument in Proposition 1 eliminates h^- beyond that halfspace. A countable-subcover argument therefore gives

$$h^- = 0 \quad \text{almost surely on } A_g^c \ .$$

Since h^- is supported on A^c, any remaining negative part is supported on $A_g \cap \{u'X > c\}$, while h^+ is supported on $A \subseteq \{u'X \le c\}$. By (17), $E_0 h^+ = E_0 h^- =: M$ and $E_0[u'Xh] = 0$. If $M > 0$, then

$$E_0[u'Xh^+] \le cM, \qquad E_0[u'Xh^-] > cM \ ,$$

so $E_0[u'Xh] < 0$, a contradiction. Hence $M = 0$, and mutual absolute continuity finishes the proof. □

Proposition 3 contains the rectangle result as a special case: take

$$A_g = \{x_1 \le c_1,\ x_2 \le c_2\}, \qquad u = e_3 \ .$$

It also covers a halfspace using only the redundant coordinate by taking $A_g = \mathbf{R}^3$.

The qualification "one halfspace" is substantive. The proposition does not show that a rectangle with an additional bad cut is admissible: the rectangle already uses the bad normal e_3, so the additional bad cut would create two bad facets. More generally, any inadmissible nice polyhedral test must require at least two nonproportional bad facets and cannot be rewritten using feasible halfspaces plus a single additional halfspace.

A minimal unresolved family is

$$A(\lambda) := \{x_1 \le \kappa,\ x_2 \le \kappa,\ x_1 + \lambda x_3 \le b,\ x_2 + \lambda x_3 \le b\}, \qquad \lambda > 1 \ , \tag{18}$$

with (κ, b) calibrated to exact size. Its two gated-face normals $(1, 0, \lambda)$ and $(0, 1, \lambda)$ are bad and nonproportional. Neither Proposition 1 nor Proposition 3 determines its admissibility. At $\lambda = 1$ the family is the admissible symmetric gate, while after appropriate rescaling it approaches a rectangle as $\lambda \to \infty$. Whether some intermediate member is inadmissible remains open. Thus the conjecture that every nice test is admissible is plausible, but is not a consequence of the results above.

6 A redundant-only test and an inadmissible example

Proposition 3 immediately handles the halfspace that uses only the redundant coordinate. We record this special case explicitly. Its admissibility does not imply that every exact-size test is admissible in the present model.

Let $z_\alpha := \Phi^{-1}(1-\alpha)$ and define

$$\phi_3(x) := I\{x_3 > z_\alpha\} \ . \tag{19}$$

Since every null mean satisfies $\mu_3 \le 0$, this test has level α, with size attained at the origin.

Proposition 4 (Redundant-only halfspace)**.** *Suppose $0 < \alpha < 1$. The test ϕ_3 in (19) is α-admissible against all level-α tests, including randomized competitors.*

Proof. Apply Proposition 3 with $A_g = \mathbf{R}^3$, $u = e_3$, and $c = z_\alpha$. □

The exact calibration in (19) matters. If the cutoff is larger than z_α, the resulting conservative test is dominated by ϕ_3; if it is smaller, the test is not level α.

There are nonetheless inadmissible exact-size tests. The following example uses all three coordinates but allocates rejection probability to evidence that the three means are equal.

Define

$$Y := \begin{pmatrix} X_3 - X_1 \\ X_3 - X_2 \end{pmatrix}, \qquad V := \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \ . \tag{20}$$

Under P_0, $Y \sim N_2(0, V)$. Let r_α denote the lower α-quantile of a χ^2_2 distribution and consider

$$\phi_{\text{eq}}(x) := I\{Y'V^{-1}Y \le r_\alpha\} \ . \tag{21}$$

Proposition 5 (An inadmissible exact-size test)**.** *The test ϕ_{eq} is a nonrandomized level-α test with size attained at the origin, but it is inadmissible against randomized competitors.*

Proof. At a general mean, let

$$\delta := \begin{pmatrix} \mu_3 - \mu_1 \\ \mu_3 - \mu_2 \end{pmatrix} \ .$$

Then $Y'V^{-1}Y$ has a noncentral χ^2_2 distribution with noncentrality parameter $\delta'V^{-1}\delta$. Its probability of falling below the fixed positive threshold r_α is at most its probability under zero noncentrality. Consequently,

$$E_\mu \phi_{\text{eq}} \le \alpha \qquad \text{for every } \mu \in \mathbf{R}^3 \ ,$$

with equality when $\delta = 0$ and strict inequality when $\delta \neq 0$. The constant randomized test $\psi(x) \equiv \alpha$ is level α and has power α at every mean. It therefore weakly dominates ϕ_{eq}, with strict inequality, for example, at $\mu = (1, 0, 0) \in \Lambda_1$. □

The test in (21) is deliberately outside the natural class considered earlier: it is not coordinatewise monotone, and its acceptance region is not convex. We do not currently have an inadmissible example among exact-size tests with closed, convex, coordinatewise-lower acceptance regions in the two-core model. The results above establish admissibility for nonnegative halfspaces, finite rectangles, feasibly supported regions, and regions obtained from feasible support by one additional halfspace. The two-bad-facet family in (18) remains unresolved.

7 One core and two redundant moments

Consider now the alternative partition

$$\mathcal{C} = \{1\}, \qquad \mathcal{R} = \{2, 3\} \ .$$

The maintained mean space becomes

$$\Lambda^{1|23} := \{\mu \in \mathbf{R}^3 : \mu_2 \leq \mu_1, \ \mu_3 \leq \mu_1\} \ . \tag{22}$$

Its null and alternative are

$$\Lambda_0^{1|23} := \Lambda^{1|23} \cap (-\infty, 0]^3, \qquad \Lambda_1^{1|23} := \{\mu \in \Lambda^{1|23} : \mu_1 > 0\} \ .$$

Unlike the two-core model, the same core coordinate is positive and dominates both redundant means at every alternative. This is exactly the condition needed by the bivariate gated transfer.

Return to the finite rectangle A_c in (10), with $P_0(A_c) = 1 - \alpha$. Fix $r \in \{2, 3\}$, and let s denote the other redundant coordinate. Choose the unique finite b_r satisfying

$$P_0\{X_1 \leq c_1, \ X_1 + X_r \leq b_r\} = \Phi(c_1)\Phi(c_r) \ . \tag{23}$$

Existence and uniqueness follow because the left side is continuous and strictly increasing in b_r, from 0 to $\Phi(c_1)$, while the right side lies strictly between these limits. Define the gated acceptance region

$$A_c^{(r)} := \{x_1 \leq c_1, \ x_1 + x_r \leq b_r, \ x_s \leq c_s\} \ . \tag{24}$$

Proposition 6 (Rectangles are inadmissible with one core)**.** *For either $r \in \{2, 3\}$, the test with acceptance region $A_c^{(r)}$ has exact size α and strictly dominates the rectangular test with acceptance region A_c over $\Lambda_1^{1|23}$. Hence every finite exact-size rectangle, including the equal-cutoff max test, is inadmissible.*

Proof. Independence and (23) give $P_0(A_c^{(r)}) = P_0(A_c) = 1 - \alpha$. Coordinatewise lowerness then gives level α over $\Lambda_0^{1|23}$.

In the $(1, r)$ coordinates, the points newly rejected by the gated test form

$$G_r := \{x_1 \leq c_1, \ x_r \leq c_r, \ x_1 + x_r > b_r\} \ ,$$

whereas the points newly accepted form

$$D_r := \{x_1 \leq c_1, \ x_r > c_r, \ x_1 + x_r \leq b_r\} \ .$$

Calibration gives $P_0(G_r) = P_0(D_r)$. This common probability is positive. Indeed, calibration forces $b_r < c_1 + c_r$: otherwise the original bivariate rectangle would be strictly contained, up to a positive-probability set, in the gated region, contradicting (23). For every $g \in G_r$ and $d \in D_r$,

$$g_1 + g_r > d_1 + d_r, \qquad -g_r > -d_r \ .$$

At any $\mu \in \Lambda_1^{1|23}$, write

$$u := \mu_1 > 0, \qquad v := \mu_1 - \mu_r \geq 0 \ .$$

The likelihood ratio in the selected coordinates is proportional to

$$\exp\{u(x_1 + x_r) + v(-x_r)\} \ .$$

For a direct strict comparison, set

$$\ell(x) := u(x_1 + x_r) - vx_r, \qquad L := ub_r - vc_r \ .$$

Then $\ell(g) > L$ for every $g \in G_r$, whereas $\ell(d) \leq L$ for every $d \in D_r$. Consequently,

$$\int_{G_r} e^{\ell(x)}\, dP_0(x) > e^L P_0(G_r) = e^L P_0(D_r) \geq \int_{D_r} e^{\ell(x)}\, dP_0(x) \ .$$

The common likelihood-ratio normalizing factor therefore gives $P_\mu(G_r) > P_\mu(D_r)$. The remaining constraint $X_s \leq c_s$ is independent and common to both tests, so multiplying by $\Phi(c_s - \mu_s) > 0$ proves strict power domination. □

Only one redundant face needs to be gated to establish inadmissibility. The test in (24) still uses all three moments: the other redundant coordinate remains through its original rectangular face. The same construction works with either redundant coordinate.

It is also easy to construct admissible tests that gate both redundant moments. For finite a, b_2, b_3, consider

$$A^{\mathrm{gg}}_{a,b_2,b_3} := \{x_1 \leq a,\ x_1 + x_2 \leq b_2,\ x_1 + x_3 \leq b_3\} \ . \tag{25}$$

The feasible-normal cone for (22) is

$$\mathcal{F}^{1|23} = \{a \geq 0 : a_2 \leq a_1,\ a_3 \leq a_1\} \ .$$

Every finite normal of (25) is a nonnegative combination of

$$e_1, \qquad (1,1,0), \qquad (1,0,1) \ ,$$

and therefore belongs to $\mathcal{F}^{1|23}$. Once the three cutoffs are calibrated to give $P_0(A^{\mathrm{gg}}_{a,b_2,b_3}) = 1 - \alpha$, the feasible-normal sufficient condition establishes admissibility.

The pairwise transfer proof does not show that a jointly calibrated double-gated test dominates the original rectangle. After the first gate is introduced, the remaining constraints no longer factor into an independent bivariate problem. A randomized mixture of the two single-gate dominators does dominate the rectangle, but deterministic domination by a double-gated test remains unresolved.